

8 bit addition :

AIM: To write an assembly language program to implement 8 bit addition using 8085 processor.

Program:

LDA 8050	(load accumulator with first number in address 8050)
MOV B, A	(move data from accumulator to 'B' register)
LDA 8051	(load accumulator with second number in address 8051).
ADD B	(add data of 'B' register with accumulator)
STA 8052	(store the data output of the accumulator in address 8052)
HLT	(Halt)

Input

Address	Data
8050	1
8051	2

Output:

Address	Data
8052	3

OK

Algorithm:

- 1) Start the program by loading the first data into the accumulator.
- 2) Move the data to a register.
- 3) Get the second data and load it into accumulator.
- 4) Add two register contents.
- 5) Check for carry.
- 6) Store the value of sum and carry in memory location.
- 7) Halt.

RESULT: Thus the program was executed successfully using 8085 processor simulator.



Registers			Flag	
A	15		S	0
BC	0A 00		Z	0
DE	00 00		AC	1
HL	00 00		P	0
PSW	00 00		C	0
PC	42 0C			
SP	FF FF			
Int-Reg	00			

Decimal - Hex Conversion

Decimal	Hex
0	0
→ To Hex	← To Dec

I/O Ports

0	-	00
Update Port Value		

Memory

8051	-	0B
Update Memory		

Load me at

```
1 LDA 8050
2 MOV B, A
3 LDA 8051
4 ADD B
5 STA 8052
6 HLT
```

Data Stack KeyPad Memory I/O Ports

Start

OK

Address (Hex)	Address	Data
1F74	8052	21
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0
1F80	8064	0
1F81	8065	0
1F82	8066	0
1F83	8067	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

